

# **Player Statistics Visualization using PCA & UMAP**

# 1. Problem Statement

Football datasets describe thousands of players through many attributes at once. Analyzing them together is difficult because player data is **high-dimensional**.

## SELECTED PLAYER ATTRIBUTES



*7 attributes per player → hard to inspect together*

**Tables & individual comparisons may not reveal:**

1

Similar player profiles

2

Groups of players

3

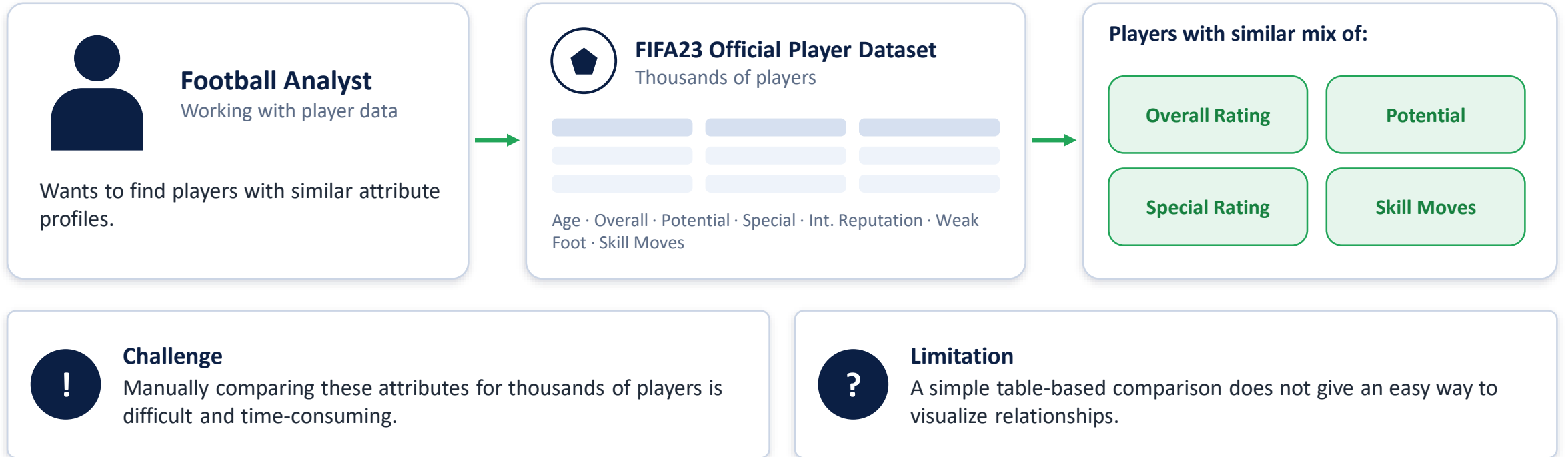
Relationships between multiple attributes

4

Hidden patterns in the dataset

**The need:** a Machine Learning-based approach that reduces the dimensionality of player data and gives an effective visual representation. Our project uses **PCA** and **UMAP** to transform multiple player attributes into a 2D representation.

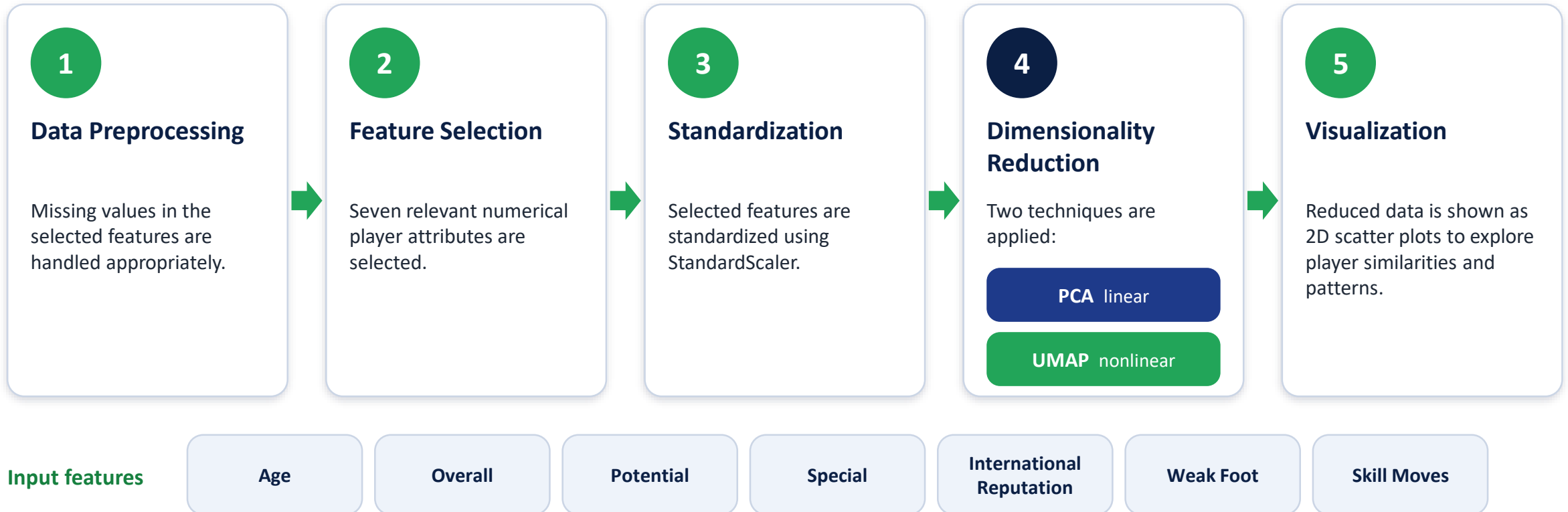
## 2. Case Scenario



**The analyst needs to answer:** *“Which players have similar attribute profiles, and what patterns exist among the players?”*

# 3. Proposed Solution

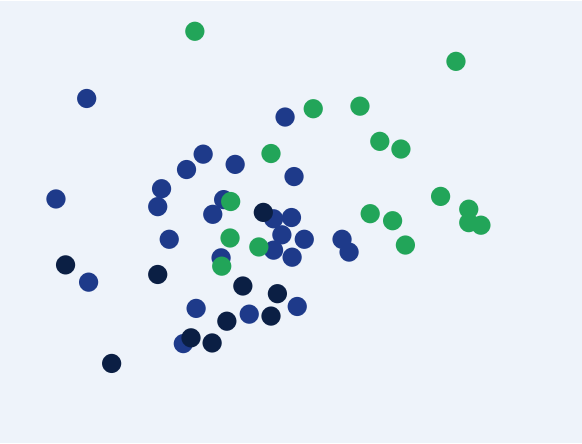
“Player Statistics Visualization Using PCA and UMAP” applies Machine Learning-based dimensionality reduction to address this problem.



## 4. Case Outcome

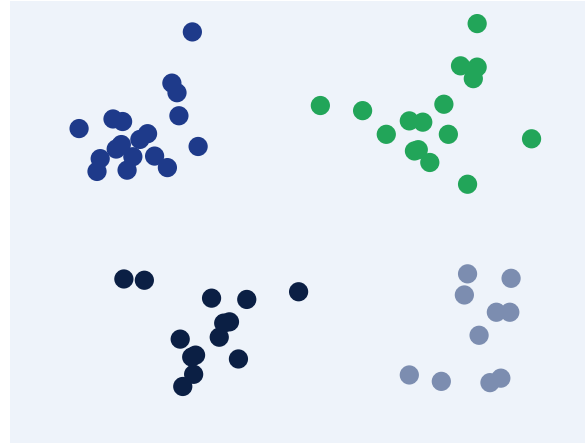
### PCA

Linear · 2D



### UMAP

Nonlinear · 2D



*Illustrative sketch: players with similar attribute patterns may appear closer together.*

The analyst can visually explore:

1

Similar player profiles

2

Groups of players

3

Patterns in player attributes

4

Relationships between players

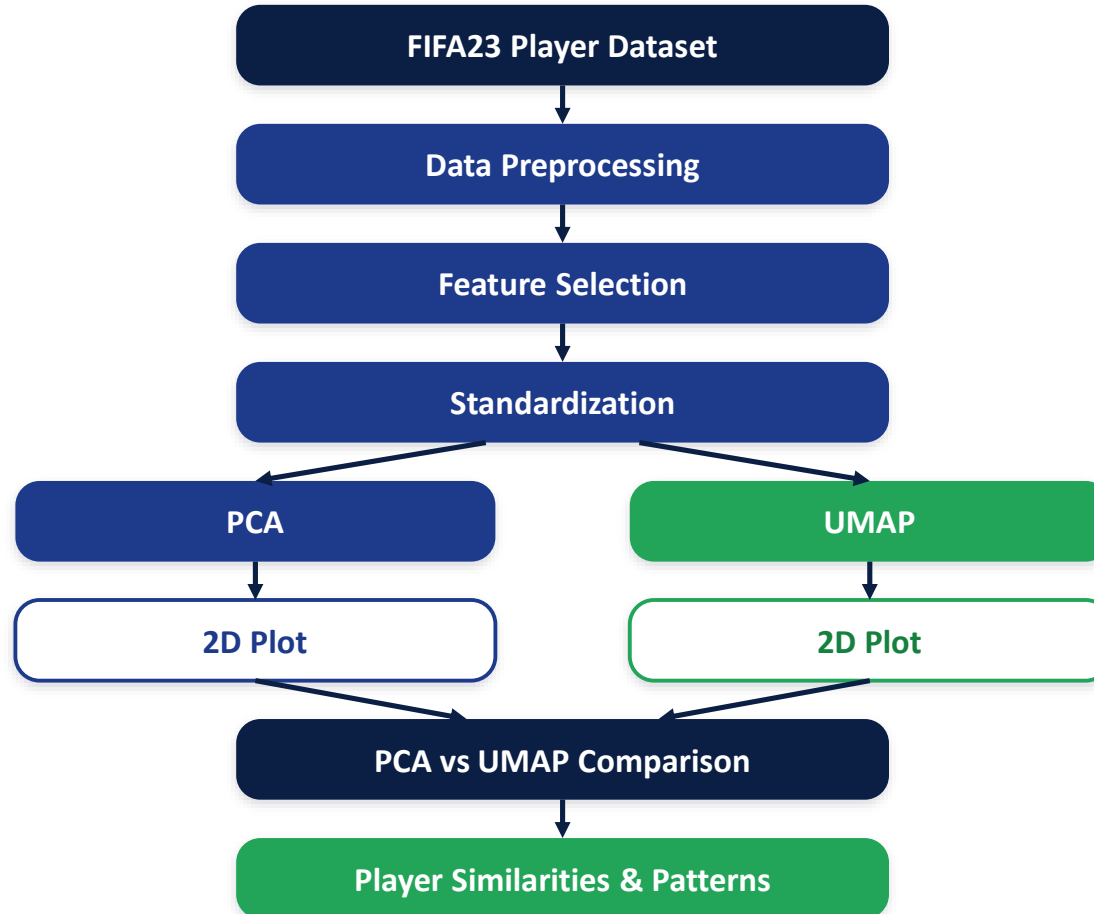
### Compare PCA and UMAP

Understand how the two techniques represent the same player dataset.

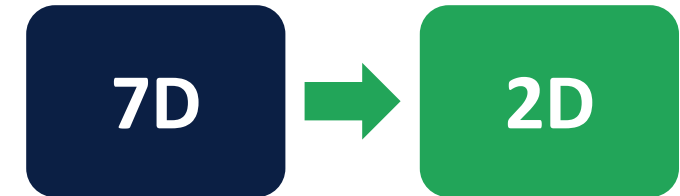
### More efficient exploration

Use visual representations instead of manually comparing thousands of player records.

# 5. Final Solution



## In short



The system converts the original seven-dimensional player attribute data into two-dimensional visual representations using PCA and UMAP.



## 6. Expected Result

1

Reduce the dimensionality of the FIFA23 player dataset.

2

Represent multiple player attributes in a 2D visualization.

3

Help identify players with similar attribute profiles.

4

Reveal groups and patterns within the player dataset.

5

Generate separate PCA and UMAP visualizations.

6

Allow comparison between PCA and UMAP representations.

7

Make multidimensional player data easier to visually explore and analyze.

**The project provides a simplified 2D visual representation of FIFA23 player attributes, helping analysts explore player similarities, groups, and patterns using PCA and UMAP.**